REFERENCE AGREEMENT IS NOT ACOUSTIC CORRECTNESS: EVALUATING LARGE AUDIO LANGUAGE MODEL CAPTIONS ON CLOTHO

Anandaswarup Vadapalli

International Institute of Information Technology, Hyderabad, India

ABSTRACT

Audio captioning systems are commonly evaluated by comparing generated captions with human references. However, such metrics may not reflect whether a caption correctly describes the audio, particularly when large audio language models (LALMs) generate details that are absent from the references. In this work, we investigate the relationship between reference agreement and acoustic correctness by evaluating four LALMs on the 1,045 recordings of the Clotho v2.1 evaluation split. We compare model captions using twelve reference-based metrics, two audio-aware metrics, and a matched human leave-one-out baseline. We further conduct a two-stage, single-reviewer assessment of 200 captions, comparing them first with the references and then with the audio. Our results show that all four models score below the human baseline on reference-based metrics, with Audio Flamingo 3 scoring highest. However, the audio-grounded assessment ranks the models differently: GPT-Audio-1.5, third or fourth on the reference-based metrics, is rated accurate for 47 of 50 recordings, and 56 of the 64 captions with weak reference agreement have no major acoustic error. Reference-based metrics alone can therefore give a misleading picture of caption correctness and model performance. We recommend combining reference-based and audio-aware metrics with audio-grounded assessment when evaluating LALM captions.

***Index Terms*—** Audio captioning, large audio language models, evaluation metrics, Clotho, human references

1. INTRODUCTION

Automated audio captioning (AAC) describes the sound sources, events and acoustic characteristics of a recording in natural language [1]. Large audio language models (LALMs) now perform this task without task-specific training, and produce longer and more detailed descriptions than captioning systems trained on AAC datasets [2–5].

AAC systems are commonly evaluated by comparing generated captions with human references using metrics such as BLEU [6], METEOR [7], CIDEr [8] and SPIDEr [9]. However, a single audio recording may contain several sound events, and human annotators do not necessarily describe all of them. Consequently, a generated caption may correctly describe an audible event that is not mentioned in the references and receive a low evaluation score. Conversely, a caption may agree with the references while incorrectly describing some aspects of the recording. Reference-based metrics cannot reliably distinguish acoustic errors or hallucinations from correct descriptions that the references do not cover, which limits their usefulness for assessing the faithfulness of LALM captions.

In this work, we investigate how well reference-based evaluation metrics reflect the acoustic correctness of LALM-generated captions. We evaluate four LALMs on the Clotho v2.1 dataset [10] and address the following research questions:

1. **RQ1: How closely do model captions agree with the five human reference captions?**
2. **RQ2: How do model captions compare with held-out human captions in metric scores and caption length?**
3. **RQ3: What patterns of agreement, additional detail, omission and mismatch appear in model captions?**

2. RELATED WORK

2.1. Automated audio captioning

AAC benchmarks such as AudioCaps [11] and Clotho [10] provide several crowdsourced captions per evaluation recording, and the DCASE challenge has made them standard [1]. In Clotho, annotators write captions, other annotators correct or rephrase them, and the five captions rated highest for accuracy and fluency are kept [12]. Each Clotho reference is a single sentence of 8–20 words, so it describes a selection of the sounds, typically the most salient, rather than the whole recording. Larger, weakly labelled collections such as WavCaps [13] support pre-training. As a result, the human references may not cover all the audible events in a recording, making it difficult to evaluate captions that include additional details.

2.2. Large audio language models

LALMs couple a pre-trained audio encoder with a large language model and follow free-form instructions about audio [2, 3, 14, 15]. Recent models are either general multimodal assistants, such as Qwen2.5-Omni [4], or audio specialists, such as Audio Flamingo 3 [5]. Unlike conventional AAC systems trained on human reference captions, LALMs are not necessarily constrained by the length or vocabulary of the benchmark captions. They may therefore add correct details that no reference mentions, but may also describe sounds that are not present in the audio [16]. Evaluating such details requires the original recording.

2.3. Audio caption evaluation metrics

Reference-based metrics compare a caption with the human references. BLEU [6] and CIDEr [8] count n-gram overlap, METEOR [7] also matches stems and synonyms, SPICE [17] compares scene graphs, and SPIDEr [9] averages CIDEr and SPICE. Zhou et al. [18] found that these metrics agree poorly with human judgements of audio captions and proposed FENSE, which adds a fluency penalty to Sentence-BERT similarity. Later metrics use contextual embeddings [19], sound descriptors [20], a language-model judge [21] or several candidates [22], but all measure similarity to the references, not to the recording. Audio-aware metrics use the recording itself: CLAP [23, 24] embeds audio and text in a joint

space, analogous to CLIPScore [25] for images, and MACE [26] combines CLAP audio–text and text–text similarity with a fluency penalty. However, both metrics measure overall audio–text similarity and do not explicitly verify individual sound events.

2.4. Reference agreement and acoustic grounding

Previous studies in image captioning have examined the limitations of reference-based evaluation using human agreement and image-grounded assessments. We draw on two such approaches in this work. First, for the COCO captions, each human caption is scored against the remaining references, which shows how well a human scores on the same metrics [27]. Second, CHAIR checks the objects a caption mentions against the objects annotated in the image, and showed that captions can score well on reference-based metrics while describing objects that are not present [28]. Previous studies of AAC evaluation have compared automatic metric scores with human preferences based on caption accuracy and fluency [18, 26]. However, the relationship between reference agreement and acoustic correctness has received relatively limited attention for LALM-generated captions. In this work, we use a matched human leave-one-out baseline to compare model and human reference agreement (RQ2). Since audio recordings have no object annotations, we also manually assess a sample of captions, including details absent from the references, against the recording (RQ3).

3. METHODOLOGY

3.1. Data and models

We use the complete evaluation split of Clotho v2.1 [10], provided by the Audio Research Group, Tampere University: 1,045 recordings of 15–30 s (6.5 h in total), each with five human captions (5,225 captions). We evaluate two API models, GPT-Audio-1.5 and Gemini 3.8 Flash, and two open-weight models, Qwen2.5-Omni-7B [4] and Audio Flamingo 3 [5], at pinned checkpoint revisions. Every model received a single user turn with the full recording followed by the same instruction (prompt template below). The models were not provided with reference captions or a prescribed caption length. We did not perform model-specific prompt tuning. The only difference in the inputs was the audio encoding required by each provider.

Prompt template (one user turn, identical for all models)
`[audio]` *full recording, 15–30 s, no cropping*
`[text]` Describe the sounds heard in this audio recording in one clear sentence that mentions the main sound sources, audible events, and distinguishing acoustic characteristics. Do not speculate about sources or events that are not supported by the audio. Respond with the sentence only, as plain text, without JSON, lists, labels or other formatting.

Decoding used temperature 0, top-p 1 and a seed of 42 where supported; Gemini 3.8 Flash ran at its lowest thinking level (`low`). This gave one caption per model and recording (4,180 in total).

3.2. Metrics and human leave-one-out calibration

3.2.1. Metrics

The twelve reference-based metrics are BLEU-1 to BLEU-4, METEOR, CIDEr-D, SPICE, SPIDEr, SPIDEr-FL, FENSE, and the raw and baseline-rescaled F1 of BERTScore, computed on PTB-tokenized text with the aac-metrics package [29]. The two audio-aware metrics are CLAP similarity, the cosine between the audio and caption embeddings of MS-CLAP 2023 [24] (each clip embedded whole as the duration-weighted mean of 7 s chunks), and MACE [26].

3.2.2. Matched human calibration

To provide a human baseline for the metric scores, we use a matched leave-one-out evaluation of the five human captions of each recording. We hold out each human caption h_j of a recording in turn and score it against the remaining four, $R_{-j} = H \setminus \{h_j\}$, where $H = \{h_1, \ldots, h_5\}$. Each model caption is scored against the same five four-reference subsets $R_{-1}, \ldots, R_{-5}$, so model and human captions face identical references. Scores are averaged over j per recording; corpus-level metrics are computed per j and averaged. This baseline measures inter-reference agreement, not caption quality or acoustic correctness. Because CIDEr-D, and hence SPIDEr and SPIDEr-FL, depend on the reference set, five-reference (RQ1) and four-reference (RQ2) scores are never compared with each other. Clotho's protocol keeps original and rephrased captions without deduplication, and 33% of held-out captions have a word-set Jaccard overlap of at least 0.6 with a reference. This threshold is operational, not a test of paraphrase; as a sensitivity analysis, we repeat RQ2 without these captions.

3.3. Caption length and qualitative review

Caption length is the number of whitespace-separated words; we relate it to each score with Spearman correlations over captions. For RQ3, we manually evaluated 200 captions: the four model captions of each of 50 recordings selected at random (seed 42), independently of the scores. The evaluation had two stages:

1. *Stage 1, reference-only.* Without audio, the reviewer compared each caption with the five references and assigned one of six reference-agreement categories: aligned (1 strong agreement, 2 mostly aligned, 3 mostly aligned with additional details), partially aligned (4 with missing details, 5 with additional details), or 6 weak agreement or source/event mismatch. Phrases that no reference mentions (*reference-unsupported phrases*) were listed.
2. *Stage 2, audio-grounded.* After the stage-1 labels were final, the reviewer listened to each recording and assigned each caption one of three ratings: (1) accurate, where omitting a secondary sound alone does not lower the rating; (2) minor errors: principal source and event correct, but a wrong attribute or order or an unsupported secondary detail; or (3) major error: principal source or event wrong or invented, or the dominant sound omitted. Each reference-unsupported phrase was judged audible, inaudible or unverifiable in the recording.

A single reviewer performed both stages by hand, blind to model identity; no AI system assigned or suggested labels.

3.4. Statistical analysis

The recording is the unit of analysis, and every comparison is paired. We test mean differences with two-sided sign-flip permutation tests (10,000 permutations) and report 95% percentile bootstrap intervals over recordings (10,000 resamples). Holm correction is applied within predefined families over all 14 metrics: 84 model-pair tests for RQ1 and 56 model-versus-human tests for RQ2. We report the paired effect size d_z, because with 1,045 recordings very small differences are significant. Ranking stability is the share of bootstrap resamples that reproduce the observed model order. To relate a metric to the audio ratings within recordings, we take every pair

of captions of the same recording with different ratings and compute the share in which the metric scores the better-rated caption higher, counting ties as one half (*concordance*; 0.5 is chance). Its intervals and tests resample or sign-flip whole recordings, with Holm correction over the ten metrics of Table 1 (not BLEU-2, BLEU-3, SPIDEr-FL or raw BERTScore). Review proportions carry Wilson 95% intervals.

4. RESULTS

4.1. RQ1: Agreement with five references

Audio Flamingo 3 scores highest and Qwen2.5-Omni-7B second on every reference-based metric, and the two API models take the third and fourth places (Table 1, left). The order is stable: at least 99% of bootstrap resamples reproduce it for nine of the twelve metrics, and only the order of the two API models varies for BLEU-3 (87%), SPIDEr and SPIDEr-FL (55%). Across all 14 metrics, 76 of the 84 model-pair differences are Holm-significant.

The audio-aware metrics rank the models differently. GPT-Audio-1.5 obtains CLAP similarity and MACE scores of 0.466 and 0.573, close to those of Qwen2.5-Omni-7B (0.462 and 0.575), and Gemini 3.8 Flash the lowest on both. Per caption, the reference-based metrics correlate with each other at Spearman ρ = 0.41–0.88, while CLAP similarity and MACE correlate 0.89 with each other but only 0.28–0.67 with the reference-based metrics. The two groups of metrics therefore measure different aspects of the captions.

4.2. RQ2: Comparison with held-out human captions

Table 1 (right) compares the model captions with the held-out human captions under the same four references. All four models score below the human baseline on every reference-based metric, and all of these differences are Holm-significant, with paired effect sizes from -0.48 to -2.55. As the baseline measures inter-reference agreement, the lower model scores indicate weaker agreement with the human references, not necessarily lower acoustic correctness.

The audio-aware metrics are the exception: on CLAP similarity, Audio Flamingo 3 is not significantly different from the human captions ($+0.006$, $d_z = 0.06$), the only one of the 56 comparisons that is not significant. Without near-paraphrase captions, the human baseline falls (CIDEr-D 0.91 to 0.46), but every model remains significantly below it on all reference-based metrics; Audio Flamingo 3 rises above it on CLAP similarity, at the uncorrected level only.

Model captions are 1.45 (Gemini 3.8 Flash) to 1.85 (GPT-Audio-1.5) times as long as human captions, and 92% of GPT-Audio-1.5's captions are longer than every reference of their recording. For the models, length is negatively associated with the overlap metrics, from -0.09 (Gemini 3.8 Flash) to -0.63 (Audio Flamingo 3) for CIDEr-D and from -0.19 to -0.52 for rescaled BERTScore. CLAP similarity and MACE are almost independent of length except for Audio Flamingo 3 (-0.11 and -0.17), and human captions show near-zero correlations throughout. However, these correlations do not establish that length causes the lower scores.

4.3. RQ3: Agreement patterns and acoustic correctness

The sample was similar to the full set on the measured summary statistics (standardized differences $\leq$ 0.37). The reference-only and audio-grounded assessments rank the four models differently (Table 2). Against the references, Audio Flamingo 3 and Qwen2.5-Omni-7B have the most aligned captions and Gemini 3.8 Flash

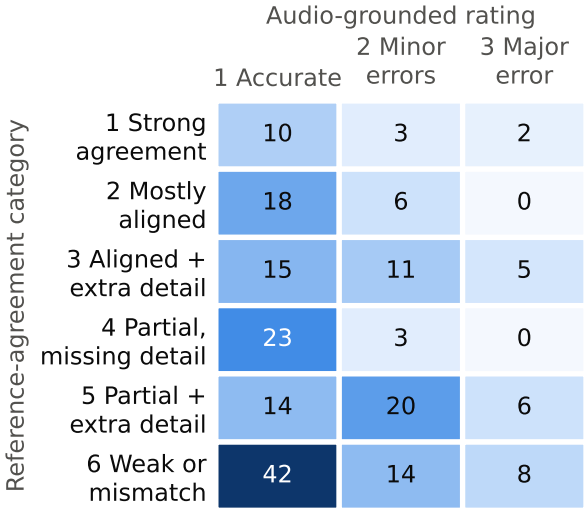


Fig. 1. Stage-1 reference-agreement category (rows; references only) by stage-2 audio-grounded rating (columns; by listening) for the 200 sampled captions of all four models, single reviewer.

the fewest, and GPT-Audio-1.5 has the most captions with missing detail. Against the audio, GPT-Audio-1.5, third or fourth on the reference-based metrics, is rated accurate for 47 of the 50 recordings (94%, 95% CI 84–98%), with no major error. The other three models are rated accurate for 23 to 27 recordings (46–54%, intervals within 33–67%) and have 5 to 9 major errors, in an unstable order. Two further analyses examine these differences:

1. *Details absent from the references.* The 200 captions contain 262 reference-unsupported phrases. Of these, 64% were audible in the recording, 28% inaudible and 8% unverifiable; these proportions are descriptive, because phrases of a caption are not independent. Inaudible phrases occur in 26 of Gemini 3.8 Flash's captions, 19 of Qwen2.5-Omni-7B's and 15 of Audio Flamingo 3's, but in only one of GPT-Audio-1.5's. At the caption level (Fig. 1), 56 of the 64 captions in category 6 had no major error (42 accurate, 14 minor errors), and 23 of the 26 category-4 captions were rated accurate. These results show that, under the review rubric, captions with weak reference agreement were often acoustically correct, possibly because the references do not describe all the audible events in a recording. Conversely, agreement did not guarantee correctness: 7 of the 70 captions in categories 1–3 had a major error, 2 of them in category 1.
2. *Metrics and the two reviews.* Across models, the metrics follow the reference-only review: Kendall's τ between each metric's model order and the reference-review order is 0.67–1.00, but with the audio-review order it is 0–0.33 for every metric except CLAP similarity (0.67). Mean scores fall from category 1 to 6 on every metric but are flat or reversed across audio ratings (rescaled BERTScore 0.29, 0.31, 0.34 for ratings 1–3). Across 156 within-recording caption pairs from 42 recordings, concordance with the audio ratings ranges from 0.39 (rescaled BERTScore) to 0.58 (MACE, 95% CI [0.46, 0.69]; CLAP similarity 0.56 [0.45, 0.66]), and none of the ten metrics differs significantly from chance after Holm correction. However, the intervals are wide, so this analysis is inconclusive.

Table 1. Mean per-recording scores. RQ1 (left): model captions against all five references; bold marks the highest model. RQ2 (right): held-out human captions against the other four references, and the paired effect size d_z of each model minus that baseline under the same references; all Holm-significant except †. Five- and four-reference scores are not comparable. Last row: mean words, and d_z of model minus mean reference length. GPT: GPT-Audio-1.5; Gemini: Gemini 3.8 Flash; Qwen: Qwen2.5-Omni-7B; AF3: Audio Flamingo 3.

	RQ1: models, five references				RQ2: human	RQ2: model − human, d_z (four ref.)			
Metric	GPT	Gemini	Qwen	AF3	(four ref.)	GPT	Gemini	Qwen	AF3
BLEU-1	0.277	0.251	0.368	**0.407**	0.640	−2.49	−2.55	−1.67	−1.40
BLEU-4	0.004	0.002	0.015	**0.030**	0.234	−1.27	−1.28	−1.18	−1.05
METEOR	0.124	0.102	0.151	**0.174**	0.319	−1.84	−2.05	−1.57	−1.32
CIDEr-D	0.048	0.064	0.164	**0.191**	0.906	−1.83	−1.75	−1.54	−1.41
SPICE	0.067	0.051	0.091	**0.117**	0.232	−1.47	−1.58	−1.19	−0.91
SPIDEr	0.057	0.058	0.127	**0.154**	0.569	−1.90	−1.84	−1.59	−1.41
FENSE	0.401	0.387	0.474	**0.510**	0.575	−1.10	−1.05	−0.73	−0.48
BERTScore-F1 (rescaled)	0.254	0.238	0.331	**0.352**	0.598	−2.40	−2.47	−1.80	−1.51
CLAP similarity	0.466	0.381	0.462	**0.509**	0.503	−0.34	−0.94	−0.42	+0.06†
MACE	0.573	0.489	0.575	**0.608**	0.635	−0.61	−1.18	−0.69	−0.34
Caption length (words)	21.0	16.4	17.0	19.0	11.3	+2.40	+1.30	+1.13	+1.16

Table 2. Two-stage review of 50 sampled captions per model. Last row: inaudible phrases among the model's reference-unsupported phrases.

	GPT	Gemini	Qwen	AF3
Stage 1: reference-agreement category				
1–3 Aligned	15	8	21	**26**
4 Partial, missing detail	10	6	5	5
5 Partial, added detail	10	12	9	9
6 Weak or mismatch	15	24	15	10
Stage 2: audio-grounded rating				
1 Accurate	**47**	23	25	27
2 Minor errors	3	18	20	16
3 Major error	0	9	5	7
Category 6 with major error	0	6	1	1
Inaudible / reference-unsupported	1/59	34/80	20/58	19/65

5. DISCUSSION

Our results highlight a limitation of reference-based evaluation for LALM captions: the models with the highest reference agreement are not necessarily those with the highest audio-grounded accuracy. This is particularly evident for GPT-Audio-1.5, which ranks third or fourth on the reference-based metrics but is rated accurate for 47 of the 50 sampled recordings. Of its captions, 10 omit details present in the references, while 55 of its 59 reference-unsupported phrases are audible. Differences in the details that models and references describe can therefore lower the scores of captions that correctly describe the recording. These single-reviewer results on 50 recordings do not establish a general superiority of GPT-Audio-1.5.

The matched human baseline also requires careful interpretation: it shows lower agreement with Clotho's references, not less correct captions. Longer captions and audible details absent from the references are associated with this gap, but neither was isolated as a cause. Besides incomplete reference coverage, reference errors, ambiguous sounds and reviewer judgement may also explain why many captions with weak reference agreement were rated accurate.

Audio-aware metrics alone may not fully address these limitations. CLAP similarity gives the model ranking closest to the audio-grounded assessment, but neither it nor MACE shows significant above-chance concordance with the audio ratings within recordings. We therefore recommend reporting audio-aware metrics alongside reference-based ones and checking sampled captions against the audio.

6. LIMITATIONS

This study has several limitations. First, the audio-grounded assessment was conducted by a single reviewer who had read the references before listening, so the ratings are an operational assessment rather than an independently established ground truth. Second, the manual evaluation covers only 50 recordings: the order of Audio Flamingo 3, Qwen2.5-Omni-7B and Gemini 3.8 Flash in it is reproduced in only 33% of bootstrap resamples, and the concordance analysis (156 caption pairs from 42 recordings) has little power, so its null result does not establish that the metrics cannot measure acoustic correctness. Finally, each model was evaluated with a single prompt and decoding configuration, the undisclosed training data of the API models may overlap with Clotho, and the findings are limited to Clotho's everyday sounds.

7. CONCLUSION

In this work, we investigated how well reference agreement reflects the acoustic correctness of captions from four LALMs on Clotho v2.1. The reference-based ranking of the models differed from the single-reviewer audio-grounded ranking, and 56 of the 64 sampled captions with weak reference agreement had no major acoustic error. These findings highlight the limitations of relying exclusively on reference agreement; we recommend complementing reference-based metrics with audio-aware measures and direct assessment against the recording, particularly when comparing models whose captions differ in length and level of detail.

Availability. Code and data: `https://github.com/anandaswarup/lalm-acoustic-caption-eval`.

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